10 (a) A cloud chamber can be used to detect α (alpha)-particles and β (beta)-particles. Alcohol in the cloud chamber exists as a vapour and condenses on ions produced in the air. This forms visible tracks.

Fig. 10.1 shows the tracks when a source of α-particles and β-particles is present in the cloud chamber.

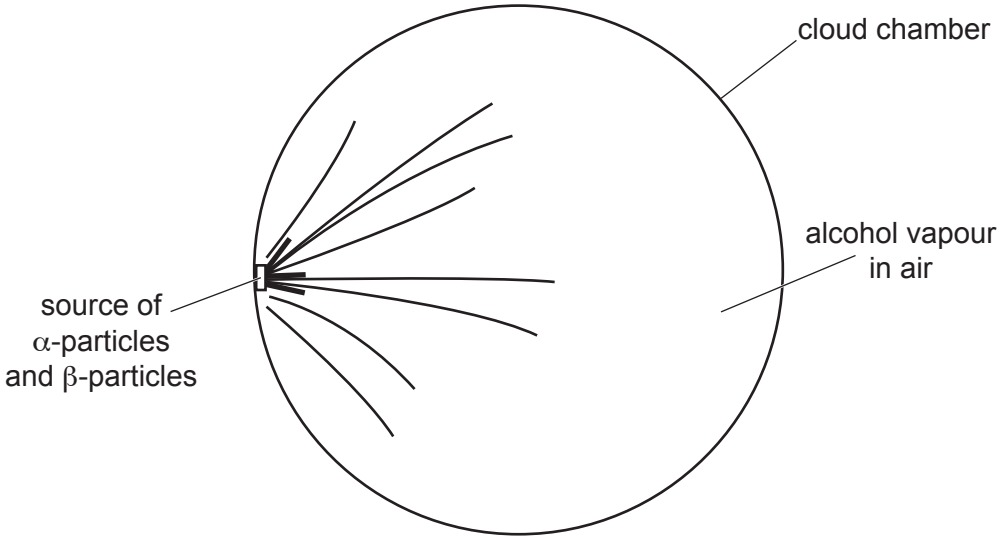


Fig. 10.1

Some of the tracks are short and thick. Other tracks are longer and thinner.

State and explain which tracks are produced by α-particles and which tracks are produced by β-particles.

α-particles .....

.....

β-particles .....

..... [3]

(b) A radioactive isotope of sodium (Na) is used to detect leaks from water pipes. A nucleus of this isotope of sodium contains 11 protons and 13 neutrons. This nucleus decays by emitting a β-particle to form a nucleus of magnesium (Mg).

(i) Describe what is meant by an isotope.

.....

.....

..... [2]

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(ii) Write down the nuclide equation for the decay of this isotope of sodium to magnesium.

[3]

(iii) This isotope of sodium has a half-life of 15 hours. The isotope of magnesium is stable and does not undergo radioactive decay.

Suggest why these properties of the isotope of sodium and the isotope of magnesium make this isotope of sodium suitable to detect leaks from water pipes.

.....

.....

..... [2]

[Total: 10]